

PBHRC Clinic Booking Data — Business Problem & Objective

A summary of the dataset scope, the operational problem it surfaces, and the analytics objective for the PBHRC_Demo_Data.xlsx workbook.

1. Dataset Overview

The workbook contains a single sheet, **Healthcare_Data_5000**, with **5,000 patient appointment records** and **27 fields** from a two-clinic outpatient network (Elgin and Canning) operating in Kolkata and South 24 Parganas, West Bengal. Each row represents one booking and captures patient demographics, booking channel, appointment scheduling details, visit outcome, wait/consultation time, revenue, and payment status.

Field group	Examples
Patient details	Patient ID, Name, Gender, Age, Phone Number
Booking details	Booking ID, Clinic, Doctor, Source, Consultation Type, Patient Type
Scheduling	Appointment Date/Time, Booking Date
Visit outcome	Status, Arrival Status, No Show, Cancelled, Rescheduled, Follow-up
Service metrics	Waiting Time (Min), Consultation Duration (Min)
Financials	Revenue, Payment Status
Location	Service, State, District

2. Key Numbers at a Glance

Metric	Value
Total bookings	5,000
No-shows	960 (19.2% of bookings)
Cancellations	1,004 (20.1% of bookings)
Reschedules	1,015 (20.3% of bookings)
Completed visits	997 (19.9% of bookings)
Bookings still open ("Booked")	1,024 (20.5% of bookings)
Average patient waiting time	≈ 44.5 minutes (completed visits)
Total recorded revenue	■ 328,100
Payment status split	Paid 1,592 · Pending 1,688 · Free 1,720

Figures are derived directly from the uploaded dataset (5,000-row extract).

3. Business Problem

Nearly **6 in 10 bookings never convert into a completed, revenue-generating visit** — roughly 19% end in a no-show, 20% are cancelled, and a further 20% are rescheduled. Combined with long average waiting times (~44.5 minutes) and an uneven payment-collection pattern (a third of visits remain "Pending"), the clinic network is losing consult capacity, doctor time, and revenue without a clear, data-driven view of *where* and *why* this is happening. Management currently lacks visibility into:

- Which clinic, doctor, service line, or booking channel drives the highest no-show and cancellation rates.
- Whether certain patient segments (age group, new vs. repeat, walk-in vs. online booking) are more prone to drop-off.
- How patient waiting time and consultation duration affect satisfaction, arrival punctuality, and repeat visits.
- How much revenue is being left uncollected (Pending/Free payment status) and where it concentrates.
- Which acquisition sources (Google, Facebook, WhatsApp, Website, Walk-in, Relative referral) yield the most reliable, paying patients.

4. Objective of the Analysis

Use the booking-level dataset to build a diagnostic and monitoring view of clinic operations that helps PBHRC reduce appointment leakage, shorten waiting times, and improve revenue capture. Specifically, the analysis should:

- Quantify and track no-show, cancellation, and reschedule rates by clinic, doctor, service, and time period.
- Identify the patient and booking-channel segments most associated with appointment drop-off.
- Measure average waiting time and consultation duration by clinic/doctor to flag scheduling bottlenecks.
- Analyze revenue and payment-status trends to highlight collection gaps and high-value service lines.
- Provide clinic managers with actionable, segment-level recommendations to improve show-up rates, patient experience, and revenue realization.

In short: turn 5,000 raw booking records into a clear picture of appointment-conversion leakage and revenue efficiency across the PBHRC clinic network — and point to where operational fixes will have the greatest impact.

Source file: PBHRC_Demo_Data.xlsx (sheet: Healthcare_Data_5000, 5,000 rows × 27 columns). This is a demo/synthetic dataset used for illustrative healthcare-operations analytics.